PRIVOKE: A Client-Side Semantic Privacy Firewall for User Prompts in Web-Hosted Large Language Models

Nicholas Bisset, Manpreet Sahadevan, Nixon Ngai, and Sanoop Mallissery

***Abstract*—Useful prompts often contain information that users may not intend to disclose to an external language-model provider. PRIVOKE inspects supported requests locally and applies explicit allow, warn or block actions before transmission. A separate training service checks version and retention constraints before distributing accepted parameters, while independently randomized telemetry provides no training data. Across 1,560 authored cases with provisional targets, updated packages improved contextual policy agreement. Targeted detector removal nevertheless exposed a tradeoff between unnecessary restrictions and missed protection, and a frozen 498-row semantic annotation-presence check yielded specificity of only 13.36 to 21.76%. Thirty-three installed receiver probes verified the declared request behavior. Every matched full-model candidate improved aggregate agreement but failed retention. Installed pretrained-model measurements isolated resident detector-activation cost, although a quality-tier probe blocked a benign prompt. Controlled telemetry tests verified delivery and accounting; separate audits identified a small excess over nominal privacy loss and substantial synthetic estimation error. Together, these findings demonstrate an inspectable protection and revision mechanism while leaving independent contextual accuracy, adaptation without lost protection, real-user monitoring usefulness and practical user benefit unresolved.**

***Index Terms*—Local differential privacy, local inference, model regression, prompt privacy, randomized telemetry, request enforcement.**

I. INTRODUCTION

WEB-HOSTED large language models (LLMs) support writing, coding and analysis by processing users' instructions and contextual material [1]–[3]. That context can extend well beyond the immediate question. A person seeking help with a report, for example, might include a colleague's details, health information or confidential organizational material. Information that helps the model complete the task can therefore also disclose something the user intended to keep private. Reported workplace incidents show how this can happen during otherwise legitimate work [4], [5]. Controls on subsequent storage or use cannot prevent the initial transfer, whereas inspection on the user's device offers an opportunity to intervene before local control is lost.

Deciding when to intervene involves more than recognizing names or identifiers. Prompts about the same subject can differ in whose information they contain, their intended recipients and the appropriateness of disclosure. Nissenbaum's account of contextual integrity likewise relates privacy to the norms governing information flows within a particular context [6]. Detector labels can represent some of these distinctions without establishing that a disclosure is authorized. The resulting decision must also affect the outgoing request: alerting a user to sensitive content does not necessarily prevent its transmission. An evaluation must therefore distinguish contextual predictions, policy decisions and what actually leaves the device.

We address these connected decisions through PRIVOKE, an implemented client-side privacy firewall for supported outgoing prompts. It combines rules, named entity recognition (NER) and semantic classification to characterize sensitivity, intended audience and information categories, then selects ALLOW, WARN or BLOCK. ALLOW and WARN release the original request; BLOCK cancels it. WARN attempts to display a notice but does not wait for acknowledgement, so the warning itself demonstrates neither consent nor prevented disclosure. This explicit request contract makes it possible to assess classification and enforcement as related, but distinct, outcomes.

Maintaining the detector raises a further difficulty: a revision can improve average accuracy while becoming incorrect on cases its predecessor handled correctly [7]. PriVoke therefore separates learning a candidate from accepting it. A server-side runtime trains on supplied labelled examples, after which the update service checks the candidate against its exact parent model and declared retention criteria. Clients can verify accepted artifacts and refresh local parameters without an application release. Reporting follows a separate path, with coarse event fields randomized on the workstation. Neither intercepted prompts nor telemetry supplies training labels, keeping maintenance and reporting independent of user traffic as a source of supervision.

Casper, Hide and Seek and ProSan establish substantial precedent for local protection and learned sanitization [8]–[11]. PriVoke contributes an implemented lifecycle and an evaluation connecting three decisions: whether a request should leave the client, whether a revised detector should replace its predecessor, and whether randomized reports can support useful monitoring. Execution records connect classifications to transmission, version-bound comparisons expose candidate gains and harms, and resource and telemetry measurements delimit cost and information release. The contribu-

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GitHub Repository Link.

tion lies in this integration and its empirical analysis; local detection, regression analysis and randomized response are established techniques.

We organize the evaluation around three questions:

1) **RQ1: Does PriVoke detect and enforce privacy-sensitive prompt policies effectively?**
2) **RQ2: Can PriVoke update its detector without sacrificing previously protected behavior?**
3) **RQ3: What systems and privacy overhead does PriVoke introduce?**

RQ1 examines the relationship between contextual prediction and enforcement, including situations in which better classification or fewer restrictions do not mean better protection. RQ2 tests whether aggregate improvements satisfy a casewise retention requirement. RQ3 measures systems costs while distinguishing report-content privacy from estimation error and contribution selection. Together, the questions establish what successful execution, training and accounting can demonstrate, and where the evidence remains limited by provisional contextual labels.

II. Related Work and Contribution

Local intervention and contextual decisions. Casper places prompt protection on the user's device through a browser extension combining rules, entity recognition and local topic analysis [8]. In the reviewed prototype, a keyboard shortcut initiates inspection; identifiers are replaced with placeholders, users can request their restoration in responses, and topic warnings require acknowledgement. Custom rules, topics and compatible replacement models make the protection configurable. PriVoke adopts the motivation for layered local inspection but implements a different request contract: supported hooks apply decisions automatically, and WARN releases unchanged content without waiting for a user response.

Hide and Seek (HaS) seeks to preserve useful interaction by hiding entities before cloud inference and recovering them in the response [10]. Its generative variant trains hiding and recovery models, whereas its label-based variant uses an entity recognizer for hiding. ProSan selects word replacements through task-importance and privacy proxies, then trains a sanitizer for local execution [11]. PromptShield approaches enterprise protection through role/policy classification [12]. Together, these systems show why detecting information, controlling its disclosure and completing the user's task warrant separate evaluation. PriVoke's labels describe its policy, but cannot by themselves validate contextual authorization.

Retaining behavior during model revision. Model-update research distinguishes improved average performance from preservation of individual predictions. Yan et al. examine *negative flips*, in which an updated image classifier becomes incorrect on a previously correct example, and propose positive-congruent training to reduce these regressions [7]. This motivates measuring both gains and losses, although image-classification findings do not establish privacy-policy correctness. PriVoke applies the distinction to publication: it compares labels and actions against the declared base and can reject an improved candidate through a configured retention guard. Linking this decision to a versioned artifact and verified client refresh separates successful optimization from acceptance for use. The evaluated connection is the contribution; the guard remains limited to its declared cases and criteria.

Privacy of operational reporting. Local randomization supports aggregate collection without directly releasing each underlying value. RAPPOR applies randomized response to client-side statistics and addresses the additional privacy cost of repeated reporting [13]. PriVoke instead combines categorical event reports with a local budget ledger, without adopting RAPPOR's memoization. Its evaluation must consequently establish both whether the implemented sampler and accounting meet the stated event-content bounds and whether the estimates are useful. We examine finite arithmetic, estimation error and contribution selection separately because a content bound protects neither participation nor network metadata. Nor do the reported fields provide corrected contextual labels or establish a need to retrain the detector.

Contribution and comparison boundaries. These strands address different outcomes: sanitization and recovery concern disclosure and continued use, negative-flip analysis concerns changed predictions, and randomization constrains released observations. RQ1 connects layered decisions to receiver-observed transmission, without claiming Casper's entity/topic, HaS's recovery-resistance or ProSan's task-utility results. RQ2 distinguishes aggregate improvement from eligibility for version-bound replacement, while RQ3 examines cost, finite content privacy, estimation and admission selection. Making these decisions jointly inspectable reveals where they diverge. Their different targets and denominators, however, provide no basis for a numerical superiority claim.

III. System Design and Privacy Boundary

A. Scope and Trust

PriVoke intervenes when personal, third-party or organizational information is about to leave the workstation in a supported prompt request. It trusts the enabled extension, local runtime and supervisor, workstation, and authenticated publisher. Protection therefore depends on that boundary: compromised or deliberately replaced page hooks can bypass inspection, and blocking a later request cannot recover information already disclosed.

The interception path recognizes prompt POST formats sent through page `fetch` and asynchronous `XMLHttpRequest` (XHR). Workers, attachments, other transports and changed endpoints need separate validation. Users may also disable protection, deliberately bypassing inspection. An optional hosted-inference configuration moves text beyond the workstation and changes the disclosure boundary; it is consequently excluded from this paper's local-inference measurements.

Fig. 1 distinguishes prompt inspection from training on supplied examples and randomized reporting. A permitting action releases the prompt to the hosted LLM, but maintenance need not collect that prompt and the collector supplies no training labels. This separation defines the information paths without establishing that the resulting decisions are appropriate.

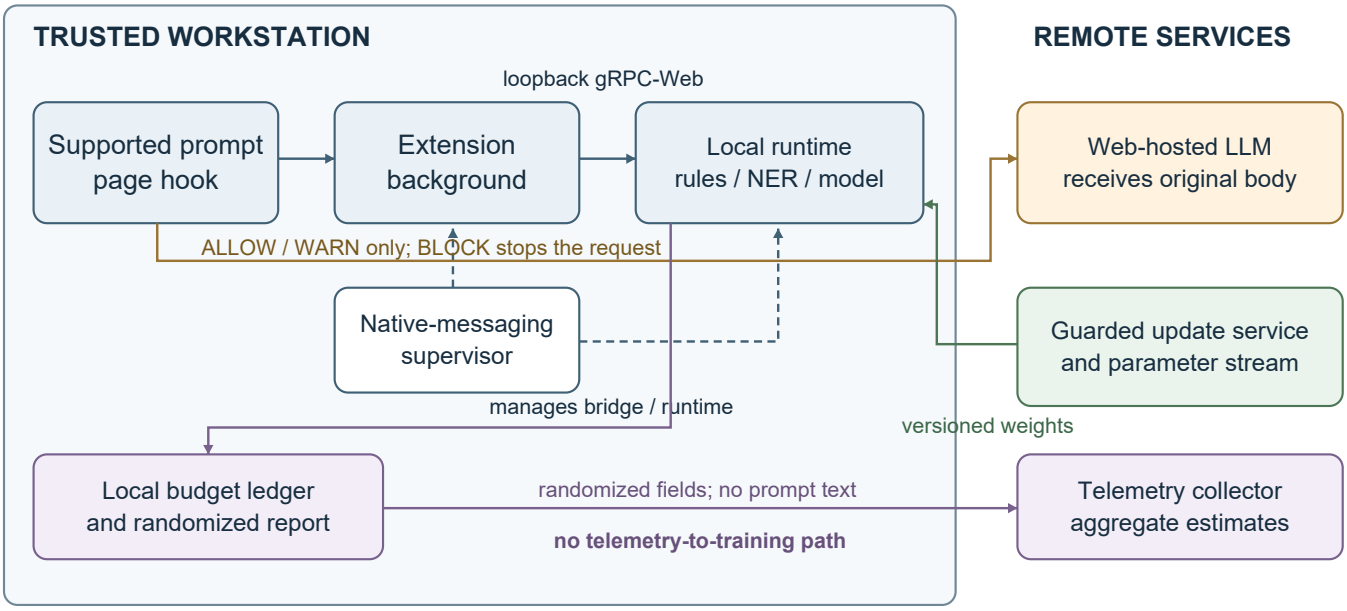


Fig. 1. PriVoke's three data paths separate local prompt inspection, supplied training examples and randomized reporting. ALLOW and WARN release the original body to the hosted LLM. Native messaging manages the bridge/runtime, while analysis uses loopback gRPC-Web. Separate services supply versioned weights and collect randomized event fields. Training uses neither intercepted prompts nor telemetry; dashed lines show lifecycle control.

B. Local Inspection and Request Enforcement

The page hook extracts supported prompt text and passes it through the extension background and loopback gRPC-Web remote-procedure-call bridge to the workstation's Python runtime. Native messaging starts the supervisor that manages this path. The returned action determines whether the original request is released or cancelled; a separate training runtime never serves as an inference fallback.

The runtime normalizes text while retaining offsets linking findings to the original input. It evaluates regular-expression rules before enabled NER and semantic layers, ending the cascade immediately if a rule returns BLOCK. Otherwise, it combines findings using the strongest action. As a result, a semantic classifier cannot reverse an earlier rule-based restriction, and an early BLOCK prevents the remaining requested detectors from running.

Failure handling also depends on which layers actually execute. A failed requested layer turns an otherwise ALLOW result into BLOCK, while valid non-ALLOW findings remain visible. The extension blocks supported requests if analysis itself is unavailable. Loss of the parameter stream can instead leave rules and NER running in a reduced configuration, which may release a request without semantic inspection. The action and execution record must therefore be interpreted together.

Semantic classification describes sensitivity, audience and information categories. Sensitivity ranges from no identified risk (S0), through low-risk personal context (S1) and potentially harmful personal information (S2), to highly sensitive information, identifiers or secrets (S3). Audience, termed *visibility* in the implementation, distinguishes public (P0), semi-public (P1), authentication-restricted (P2), group-private (P3), personal-private (P4) and unspecified (PU) contexts. Categories may overlap and cover health, politics, religion, criminal, financial, sexual, child, location, identity and third-party information.

The policy ordinarily maps S3 to BLOCK and S2 to WARN. S0/S1 allow unless identity and location occur together, or either occurs with visibility P2–P4; these exceptions produce WARN. S0 without categories always allows. Numeric confidence below 0.5 overrides this mapping: S3 becomes WARN, S2 remains WARN only when identity or location is present, and other cases become ALLOW. Confidence summarizes sensitivity and category heads rather than a calibrated probability of joint correctness or authorized disclosure. Neither the labels nor their deterministic mapping has been independently validated for contextual authorization.

Consider a hypothetical S2 prediction with HEALTH, confidence 0.7 and no earlier restriction. The policy returns WARN, allowing the supported request to send its original body without acknowledgement. If a rule had already returned BLOCK, the request would instead be cancelled before semantic inspection. This example illustrates the mapping; it adds no observed prediction, independently labelled case or test.

These models classify prompts rather than generate conversational responses. Historical V1 packages use randomly initialized frozen encoders with fitted heads; V3 packages retain pretrained Bidirectional Encoder Representations from Transformers (BERT) encoders and poolers [14]–[17]. Both families offer efficient, balanced and quality packages, whose names imply no guaranteed performance ordering. The installed default now selects V3 balanced, alongside explicit V3 tiers and labelled older V1 choices. Its CPU loader verifies

binary packages from the model service without falling back to a random or V1 model. Differences in tokenizers, capacity, thresholds and training history mean that comparisons assess the complete package change, not pretraining or architecture alone.

C. Guarded Detector Updates

Before accepting a candidate, the update path checks both its model identity and its behavior. A controller prepares bounded labelled batches and held-out guard cases, excluding protected rows, texts and groups from training. The server loads the specified base, fits a candidate, and returns parameter changes and metrics bound to that base. The updater then checks parent identity, permitted tensors and shapes, finite values, magnitude limits and configured guards. Artifact checks establish that a candidate is well formed; retention checks assess its behavior on declared cases. Neither successful fitting nor parameter transfer establishes acceptance.

Publication uses atomic compare-and-swap: the current base must still match the candidate's declared parent. This prevents a decision about one model from replacing an intervening revision. Durable replay receipts connect identical retries to their earlier artifact outcomes, avoiding duplicate application. Head and underlying-model stages are separate publications, however, so rejection of a later stage does not undo an accepted head update. Clients verify delivered artifacts before refreshing cached weights. Standard V3 instead uses a pinned read-only package store and a configuration-owned alias, beyond the separately enabled legacy training/publication route. Unsupported V3 legacy-gradient calls are rejected; installed V3 inference thus provides no evidence of deployed V3 learning or guarded publication.

Training is centralized and uses supplied examples, without user gradients or federated learning. The implemented service path establishes neither cloud deployment nor the value of a particular revision.

D. Separate Randomized Telemetry

Telemetry reports five coarse event fields: action, an action/sensitivity risk bucket, one category chosen by fixed precedence, a bounded release label and a four-hour time-of-day bin without a date. Their domains have sizes $K_i = 3, 4, 11, 2, 6$, including a no-category value. Each field is randomized locally; raw prompts, identifiers, exact confidence and per-layer findings are omitted. This reduces disclosure without making the remaining fields harmless. The collector estimates lifetime marginals for usage and development inspection, but cannot estimate latency or error rates: it has no analysis/user denominator, date/window query or joint-event record, and distinguishes only v0.3.0 from OTHER. A separate Health call checks store writability. Neither channel produces corrected training labels.

Local differential privacy limits how much an output probability can vary between possible event contents [18]. This comparison fixes the reporting opportunity and public configuration, leaving the presence of a report visible. The event budget ε is divided equally across fields as $\varepsilon_i = \varepsilon/5$. For a domain of size K_i, the true value has reporting probability p_i and each alternative has probability q_i:

$$p_i = \frac{e^{\varepsilon_i}}{e^{\varepsilon_i} + K_i - 1}, \quad q_i = \frac{1}{e^{\varepsilon_i} + K_i - 1}. \tag{1}$$

Under independent ideal sampling, composition multiplies the field likelihood ratios to bound the event ratio by e^{ε} [18]. The implementation compares a fresh uniform 53-bit integer draw, $J/2^{53}$, with a binary64 threshold $\widetilde{p}_i$, choosing alternatives by rejection sampling. The resulting finite probabilities and event loss are

$$\begin{aligned} P_i &= \frac{\lceil 2^{53}\widetilde{p}_i \rceil}{2^{53}}, \quad Q_i = \frac{1 - P_i}{K_i - 1}, \\ \varepsilon_{\text{finite}} &= \sum_i \log \max\{P_i/Q_i, Q_i/P_i\}. \end{aligned} \tag{2}$$

The calculation depends on host arithmetic and independent uniform random bits, so the ideal formula alone cannot certify the sampler. A persistent ledger reserves budget before report construction, defaulting to one per event and eight per Coordinated Universal Time (UTC) day. Failed construction or delivery does not refund that reservation; accounting therefore requires retained state and a trusted clock. Report presence, contribution selection, delivery and network metadata fall outside the content guarantee. The collector estimates marginals by inverting nominal response probabilities and clipping counts to their feasible range, leaving their usefulness after randomization and clipping to empirical assessment.

TABLE I
EVIDENCE LINKING THE RESEARCH QUESTIONS TO THEIR MEASURED OUTCOMES. CONTEXTUAL TARGETS ARE PROVISIONAL AND DO NOT CONSTITUTE INDEPENDENTLY ADJUDICATED PRIVACY TRUTH.

RQ	Evidence	What it establishes
1	1,560 contextual cases; targeted ablation; 498 final presence rows	Agreement and detector contribution on declared targets.
1	33 installed receiver probes	Request disposition on tested paths.
2	Matched head/full continuation; 16-case guard	Paired gains/harms and conditional acceptance.
3	Historical cost; installed V3 activation; telemetry audit, simulation and operation	Bounded cost, delivery/accounting and content-privacy/estimation limits.

IV. Evaluation Method

A. Questions and Evidence Units

Table I links the three research questions to evidence about prediction and enforcement, retained behavior, and systems and reporting costs. Because several analyses reuse cases, agreement between their findings does not constitute independent confirmation.

Product calls request rules, NER and semantics [2,3,4], recording any legitimate skips after a rule-based BLOCK. Sole-semantic calls explicitly request [4] and assert that it alone executes successfully. This distinction prevents product

outcomes from being attributed to semantic classification alone.

Contextual targets and measures. The authored workload contains 1,560 cases, comprising 810 sensitive and 750 clean examples, with 260 per audience class, 35 scenario families and 24 audience-expression groups. Targets for sensitivity, audience, categories and allowed actions were prepared provisionally with assistant support, without independent human adjudication. Cases sealed for an earlier semantic comparison were subsequently opened and reused for the product comparison and ablation. These remain development results on authored material, even when a later protocol was fixed prospectively.

A positive binary prediction requires nonzero sensitivity or a nonempty category set. Label-joint agreement requires exact matches for sensitivity, audience and category set; the *label-and-action joint* measure additionally requires an allowed action. We also measure allowed-action agreement separately. An action below the allowed set is under-restrictive, while one above it is over-restrictive. Receiver arrival provides a different observation: it establishes transmission rather than classification or policy agreement.

Final annotation target. The once-only final endpoint comprises 498 English rows from the Personally Identifiable Information Masking Benchmark (PIIMB), revision `4a13e9f`: 236 annotation-positive and 262 annotation-negative [19]. Source inputs, all six packages and scoring were frozen before opening, with no subsequent tuning, reselection or promotion. Sole-semantic execution evaluates the same nonzero-sensitivity-or-category projection against annotation presence. This target differs from contextual authorization and PIIMB's official character-masking/F2 task. The 464 observed source-group keys capture declared relationships, without proving template independence.

For contextual uncertainty, we use 2,000 paired whole-cluster draws with seed 42 and linear 95% percentile ranges, separately grouping by family and audience expression. Ablation expression groups were reconciled after collection. Final ranges use 2,000 paired source-group draws with the same seed and percentiles, retaining row weighting. These unadjusted ranges show sensitivity to known dependence; they establish neither crossed independence, template separation nor population coverage. Repeated calls and seeds likewise add no independent situations.

B. Comparison and Retention Protocols

Historical and updated efficient, balanced and quality packages processed the same contextual cases through the product handler, with rules and policy fixed. Updated packages descend from frozen-encoder head fitting, joint encoder/pooler/head continuation and audience-supervised rehearsal; checkpoints were selected on validation alone before the sealed comparison. Historical packages used 64/96/128-token windows and category threshold 0.38, compared with 512 WordPieces and threshold 0.5 for updated packages. Their respective parameter counts were 0.019/0.037/0.054 and 4.785/9.596/17.494 million. These differences prevent attribution to pretraining or architecture alone, while repeatability passes add no cases.

The ablation holds the updated balanced package, inputs and policy constant across full [2,3,4], no-rules [3,4] and no-NER [2,4] configurations. Execution checks distinguish requested layers from legitimate skips, allowing detector contributions on this workload to be examined without assuming that removal is safe elsewhere.

RQ2 holds the initial model and training exposure constant. Both head-only and full-model fits start from byte-identical official pretrained balanced encoder/pooler weights with zero task heads, rather than either RQ1 package. Training used Python 3.11, Torch 2.12/CUDA 13.0 and Transformers 4.57.6 on an RTX 5080. Seeds 42/43/44 received the same ordered 4,096 exposures over 64 steps, minimizing sensitivity and audience cross-entropy plus mean category binary cross-entropy with dropout disabled. AdamW used head/encoder rates $10^{-3}/2 \times 10^{-5}$, zero weight decay, norm-one clipping and microbatch 4/effective batch 64. Enabling encoder optimization in full-model training also changed the scope of joint-gradient clipping.

Excluding all declared training/guard groups and normalized texts removed 480 validation rows, leaving 1,080 reused provisional cases across 25 families: 570 sensitive and 510 clean. Each terminal candidate and the unchanged base underwent native in-process sole-semantic assessment; assessment scores selected no checkpoint. Sixteen separate guard cases required improved joint correctness and zero action harms against one derived target action per case, whereas the broader assessment accepted declared action sets. Guard acceptance and assessment improvement thus answer different questions. No candidate was published: effective-artifact mappings show what the rule would retain, conditional on the small guard and weak zero-head control.

C. Enforcement, Cost and Telemetry Protocols

Thirty-three installed Linux Chromium probes exercised the built extension, standard native installer, supervisor, bridge and historical balanced detector. A local Hypertext Transfer Protocol Secure (HTTPS) receiver checked original bodies and late sends during forwarding, blocking, faults, cancellation and recovery. Product and sole-semantic probes outside the resource windows asserted actual layer execution. These synthetic receiver observations do not establish authenticated-provider or cross-platform compatibility.

Historical cost measurements used three fixed synthetic cases, one per action, over fetch and XHR. Two fresh profiles per transport/action cell yielded twelve profiles, each completing a first decision, five warmups and thirty measured rules/NER requests, for 360 measured decisions. Page latency ran from analysis request to result, excluding receiver response; nearest-rank percentiles pool only two profile clusters per cell. Resource windows began at browser launch and ended after request/readiness checks, before teardown. Whole-container central processing unit (CPU) time and sampled charged memory include infrastructure, cache, driver, receiver and observer. Charged memory differs from process resident

set size (RSS) and proportional set size (PSS) [20]. Sampling requested 100-ms intervals and recorded actual gaps. The historical V1 study included neither semantics, an unprotected baseline nor per-role resource attribution.

The V3 study used two fresh Linux Chromium profiles per transport and the synthetic prompt "Please explain the steps in this process." Each profile retained its normal launcher, supervisor, bridge and model service across detectors-off, rules/NER, semantic and full arms, with order reversed in the second replicate. Five warmups and thirty measurements per arm produced 80 warmups and 480 timed requests. Because every timed request forwarded, request-to-receiver completion could be compared through within-profile median subtraction. The off baseline retained interception and infrastructure. Exact layers and packages were asserted on Torch 2.11 CPU with four threads, Python 3.12.3 and Chromium 153.0.8010.12; telemetry was disabled. Ten untimed post-cost probes checked the three V3 tiers, older V1 balanced and return to default over both transports.

The telemetry audit enumerated exact rational kernels on a fingerprinted Windows CPython 3.11.9 build, enclosing logarithms with outward 100-digit arithmetic under independent uniform operating-system (OS) bits. Budgets 0.5, 1 and 2 cover the declared settings. Estimator simulation crossed these budgets with $n = 8, 80, 800, 8000$, all five field domains, and balanced-integer, point-mass and minimum-one 1%-target histograms. Each of the resulting 180 prespecified cells had 10,000 repetitions with seed 20261014. NumPy 1.26.4/PCG64 sampled binary64 approximations of finite probabilities; the collector applied nominal inversion and independent $[0, n]$ clipping. Large n pools installation-days, still limited to eight default reservations each. The estimand is a fixed contributor histogram, excluding population-sampling variation, and no deployment-usefulness threshold was specified.

Ledger checks examined persistence, concurrent admission and failure accounting. A separate controlled Windows host study exercised the actual reporter, randomizer and ledger with a separate SQLite-backed collector, using generated analysis-shaped fixtures without a detector. Twenty-three ledger fixtures covered caps, reopening, ordered inputs, outage/recovery and malformed packets. Synchronous report-call timing excluded background delivery and was kept separate from Linux browser timing. Neither study involved real-user traffic.

Artifact availability. Tracked recipes in `evaluation/` and dated records in `paper/research/` identify configurations through source, model and data fingerprints and environment receipts. They distinguish product/ablation dependencies, including historical uncommitted sources, from matched continuation, final semantics, historical and standard-V3 client captures, and telemetry. Some pinned checkpoints and raw run artifacts remain local and ignored. Exact reruns therefore require more than a Git checkout, and public distribution of those assets has not been established.

V. RQ1: Detection and Enforcement

A. Contextual Policy Agreement

Updated packages more often matched both the authored labels and an allowed action. Label-and-action joint agreement rose by 27.37–41.86 percentage points to 693–879 of 1,560 cases (Table II), with no failed calls or layer errors in any of the six runs. Efficient, balanced and quality gained 484/573/671 previously incorrect joint outcomes while losing 57/16/18 previously correct ones. Allowed-action gains were 389/389/348, against losses of 3/6/0. Quality thus preserved every previous allowed-action match while losing eighteen joint matches, illustrating that an intervention can be retained even when its classification changes. These gains concern reused provisional targets; no package was promoted.

TABLE II
Full-product comparison on the same provisional contextual cases, shown as historical/updated counts. Labels+action is label-and-action joint agreement and Action is allowed-action agreement, both out of 1,560. Sensitive and Clean count correct classifications out of 810 and 750. All arms completed without errors.

Package	Labels+action	Action	Sensitive	Clean
Efficient	266/693	1029/1415	544/808	494/629
Balanced	160/717	1026/1409	771/802	318/631
Quality	226/879	1073/1421	598/810	468/641

TABLE III
Balanced-package ablation across the same 1,560 provisional cases, with requested layers in brackets. Action and Joint count allowed-action and label-and-action joint agreement; Under and Over count actions below and above the allowed set.

Configuration	Action	Joint	Under	Over
Full [2,3,4]	1409	717	12	139
No rules [3,4]	1530	972	30	0
No NER [2,4]	1409	717	12	139

Execution records identify a shared source of unnecessary restriction. All updated packages over-restricted the same 139 cases: rules blocked 109 clean controls before later detectors could run and warned on thirty low-risk controls. Rules ran on every case, ending the cascade on 289 and leaving NER and semantics to process 1,271. A better downstream semantic model cannot undo these earlier decisions. NER produced zero converted PriVoke findings after supported-label filtering, although this does not mean that the recognizer was inactive or returned no raw entities.

The fixed-package ablation examines this explanation by comparing agreement with under- and over-restriction (Table III). Removing rules corrected all 139 shared unnecessary restrictions but lost eighteen previously correct actions. Its net gain of 121 actions therefore came at the cost of missed protection under the provisional policy. Paired action-rate gains ranged from 3.46–12.70 percentage points by family and 5.96–9.23 by audience expression. These separate descriptive ranges support the direction of the tradeoff without establishing that its losses are acceptable. No-NER outputs equalled the full product, consistent with zero converted findings on this fixture. Verified execution supports that observation, but neither ablation establishes that a detector is unnecessary elsewhere.

TABLE IV
FINAL SEMANTIC ANNOTATION-PRESENCE RESULTS FOR 236 POSITIVE AND 262 NEGATIVE ROWS. HISTORICAL/UPDATED COUNTS SHOW TRUE/FALSE POSITIVES (TP/FP) AND TRUE/FALSE NEGATIVES (TN/FN). ALL SIX ARMS COMPLETED WITHOUT ERRORS.

Package	TP	TN	FP	FN
Efficient	140/218	61/45	201/217	96/18
Balanced	141/221	65/35	197/227	95/15
Quality	167/190	51/57	211/205	69/46

B. Agreement on a Separate Annotation Target

The final endpoint asked whether the packages distinguished positive from negative rows under a different annotation target. All 2,988 semantic-only calls completed without errors and verified exclusive execution of the requested layer. Updated recall reached 80.51–93.64%, but specificity remained at only 13.36–21.76% (Table IV). Efficient and balanced lost specificity relative to their historical versions, while quality's small increase remained uncertain. Paired net-accuracy gain ranges from frozen source-group resampling were nevertheless positive: 7.41–17.37, 5.35–15.03 and 1.02–10.71 percentage points, respectively. Recognizing additional positives improved average accuracy despite poor discrimination of annotation-negative rows.

This endpoint complements rather than reproduces the contextual comparison. It assesses source annotation presence through semantics alone, whereas the product comparison assesses provisional contextual labels and actions through a cascade. The difference between their results is therefore not a controlled estimate of distribution shift or detector contribution. Low final specificity still demonstrates poor discrimination under the declared projection, but neither endpoint independently establishes whether disclosure is authorized.

C. Does a Decision Control Transmission?

Across thirty-three correctness probes, the installed-browser capture connected decisions to eighteen unchanged-body sends and fifteen zero-send outcomes. Correlated runtime and receiver records, including late-arrival checks, established these outcomes without a decision broker. Healthy full-pipeline and sole-semantic probes also verified execution, linking request disposition to the detector path actually exercised.

Fault probes exposed the difference between keeping a request path available and preserving its protection. When the parameter stream was interrupted, semantic-only requests were blocked, but configured full-mode fallback ran rules and NER and forwarded the request. Restarting restored full execution; disabling protection or all layers forwarded by design. The fallback thus maintained reduced analysis without demonstrating equivalent semantic protection. These completed observations were kept separate from nine earlier failed captures.

Answer to RQ1. PriVoke improved agreement with provisional contextual policy and enforced its declared request contract on the tested installed paths. Action errors, unchanged-content forwarding under WARN and weak specificity on the separate annotation-presence task constrain the broader effectiveness claim. The results do not yet establish independent contextual accuracy, provider compatibility, prevention for every sensitive prompt or practical usability.

TABLE V
MATCHED SEMANTIC-ONLY CONTINUATION ON 1,080 REUSED PROVISIONAL CASES, SHOWN AS HEAD-ONLY/FULL-MODEL COUNTS. LABELS+ACTION DENOTES LABEL-AND-ACTION JOINT AGREEMENT. THE UNCHANGED ZERO-HEAD BASE HAS ZERO JOINT MATCHES AND 480 ALLOWED-ACTION MATCHES. GUARD HARMS ARE RELATIVE TO THAT BASE ON SIXTEEN SEPARATE CASES.

Seed	Labels+action	Allowed action	Guard harms
42	192/257	908/993	0/1
43	204/247	906/989	1/1
44	198/268	912/1003	0/1

VI. RQ2: UPDATES AND PREVIOUSLY PROTECTED BEHAVIOR

A. Improvement Subject to Retention

A higher score does not necessarily make a candidate suitable for replacement. Whereas RQ1 changes capacity, tokenizer, threshold and supervision together, RQ2 fixes the base and exposure before enabling encoder optimization. This matched comparison separates the observed learning gain from the decision to permit publication.

B. Matched Continuation and Guard Decisions

All six 64-step fits completed with finite parameters. HEADS preserved the encoder tensors, while FULL changed 39 and produced nonzero gradients in later steps. The six candidates and unchanged base completed 7,560 error-free native in-process semantic assessments, each covering the same 1,080 situations. These repeated assessments provide neither additional independent cases nor measurements of deployed network execution.

FULL improved joint and action agreement in every seed (Table V). For seeds 42/43/44, joint gains and losses were 79/14, 69/26 and 90/20; action gains and losses were 123/38, 124/41 and 110/19. Under-enforcement distance increased on 33/37/15 cases and over-enforcement distance on 17/13/17. These directional counts do not form a mutually exclusive harm union and cannot be summed as distinct harmed cases. Aggregate improvement therefore coexisted with lost protection on reused cases, without establishing how frequently either would occur in a population.

Each FULL candidate introduced one action harm relative to the unchanged base on the sixteen guard cases and was rejected. HEADS42 and HEADS44 passed, whereas HEADS43 failed. Mapping a rejected candidate back to the base forfeits its assessment gains, but these mappings remained hypothetical: no candidate in this experiment was published or promoted. Passing a small guard cannot rule out other harms or compensate for the weak zero-head baseline.

Answer to RQ2. Joint continuation of the pretrained encoder and task heads improved aggregate agreement over matched head-only training in all three seeds. Nevertheless,

TABLE VI
PAGE-OBSERVED RULES/NER DECISION LATENCY (MS), WITH SIXTY MEASUREMENTS ACROSS TWO FRESH PROFILES PER CELL. MAX IS THE OBSERVED MAXIMUM; TIMINGS EXCLUDE RECEIVER RESPONSE AND SEMANTIC PROCESSING.

Transport	Action	p50	p95	Max
fetch	ALLOW	35.0	62.5	73.6
fetch	WARN	31.9	36.1	59.4
fetch	BLOCK	27.9	31.6	33.2
XHR	ALLOW	30.8	35.8	57.7
XHR	WARN	31.3	34.6	59.2
XHR	BLOCK	28.5	32.2	34.1

every FULL candidate, together with head-only seed 43, failed retention. The experiment demonstrates bounded learning gains and an enforceable guard, but those gains did not meet the retention requirements for publication. Repeated narrow supervision may limit generalization, but the study does not isolate overfitting as the cause of its errors.

VII. RQ3: SYSTEMS AND PRIVACY OVERHEAD

A. Measured Systems Cost

Historical V1 timings in Table VI end at the local rules/NER decision. All 240 ALLOW/WARN decisions ran both detectors, whereas the 120 BLOCK decisions stopped after rules, legitimately skipping NER. The cells therefore include different amounts of work. Medians span 27.9–35.0 ms, and each maximum describes sixty requests across two profiles, excluding receiver completion and semantic processing.

Across 126.40 seconds of resource windows, the container accumulated 193.29 CPU-seconds, including 0.69 observer CPU-seconds; CPU time sums concurrent processor use. Maximum charged memory reached 1,267.91 MiB for services, cache, driver and receiver together. Sampling gaps of up to 622.84 ms could miss shorter spikes. Final-read and cleanup checks passed, but the combined resource totals cannot isolate incremental overhead.

Installed V3 balanced timings appear in Table VII. The four within-profile median increments per arm span 26.1–26.7 ms for rules/NER, 9.4–17.0 ms for semantics and 31.9–33.5 ms for full detection; these compare profile medians rather than individual requests. All 480 timed attempts returned ALLOW. Including 80 warmups, 144 scaffold calls and ten post-cost checks, the 714 calls yielded 574 analysis RPCs and 712 unchanged-body arrivals without execution errors. Quality blocked the same benign prompt once on each transport, accounting for both non-arrivals. Correct package loading thus coexisted with an inappropriate decision, although one repeated prompt cannot establish a false-positive rate.

These comparisons isolate detector activation for one short input, excluding total overhead relative to an absent extension and the cost of cold download or loading. With only two profile clusters per cell and observed maxima up to 341.0 ms, they establish neither a service-level objective nor a deployment-tail guarantee. Product code, dependencies and images also differ between V1 and V3, preventing attribution of cross-study speed differences to the model alone.

TABLE VII
INSTALLED V3 BALANCED REQUEST-TO-RECEIVER LATENCY (MS), WITH SIXTY TIMED REQUESTS ACROSS TWO PROFILES PER CELL. OFF RETAINS EXTENSION/NATIVE INFRASTRUCTURE, AND OBSERVED MAXIMA DO NOT GUARANTEE TAIL PERFORMANCE.

Transport	Detectors	p50	p95	Max
fetch	Off	3.2	4.0	4.3
fetch	Rules/NER	29.6	33.8	92.4
fetch	Semantic	19.7	158.8	283.6
fetch	Full	35.4	172.8	202.1
XHR	Off	3.3	5.0	7.4
XHR	Rules/NER	29.9	31.8	100.5
XHR	Semantic	15.1	162.8	312.7
XHR	Full	35.9	175.8	341.0

TABLE VIII
TWO ILLUSTRATIVE PRESPECIFIED CATEGORY CELLS ($K = 11$, EVENT $\varepsilon = 1$), EACH REPEATED 10,000 TIMES. TRUTH, CLIPPED BIAS AND RMSE DESCRIBE THE FIRST CATEGORY IN COUNT UNITS. CLIP GIVES ITS CLIPPING FREQUENCY; SUM GIVES THE FREQUENCY OF INDEPENDENTLY CLIPPED MARGINALS FAILING TO SUM TO n (TOLERANCE 10^{-9}).

n	Truth	Bias	RMSE	Clip (%)	Sum (%)
80	7	+22.84	42.32	85.62	99.99
8000	727	+237.50	1028.47	28.32	99.78

B. Privacy Cost and Telemetry Utility

The finite audit qualifies the privacy guarantee suggested by the ideal formula. On the fingerprinted host, maximum event loss exceeded nominal budgets 0.5, 1 and 2 by approximately $(1.294, 1.503, 1.226) \times 10^{-15}$. Although tiny, this excess means the kernel does not satisfy the exact nominal bound. Eight default reservations compose to approximately $8 + 1.202 \times 10^{-14}$ under the same independent uniform-entropy assumptions. The result bounds admitted event content under the audited conditions, without certifying other builds or deployment entropy.

The two prespecified balanced-integer category cells in Table VIII, at event budget one and $n = 80/8000$, illustrate clipping bias and root-mean-square error (RMSE). They do not establish utility acceptance or describe every cell. Absolute error grows with contributor count even when relative error can decline, and independently estimated marginals almost never sum to the total. Additional reports alone therefore do not ensure reliable counts; acceptable application error and deployment accuracy remain unmeasured.

Accounting checks admitted eight default reservations, denied the ninth, preserved the cap after reopening and reset it on the next UTC day. Clock rollback and corrupt state failed closed; sixteen of forty concurrent epsilon-0.5 requests were admitted. Construction, queue and delivery failures still consumed reservations. These checks support bounded admission under the tested state and clock conditions, but admission also determines which events are represented. In a constructed 32-event sequence containing 25% BLOCK overall, reversing the order changed BLOCK representation among the first eight contributors from 100% to 0%. Content randomization cannot correct that selection or conceal participation, so a

content-privacy bound alone cannot establish representative monitoring.

The controlled operational run offered 308 opportunities across 23 ledgers, producing 184 reservations, 168 accepted reports, sixteen submission failures and 124 pre-queue suppressions. Reopen, recovery and invalid-packet checks passed, and noisy and estimated summaries survived collector restarts. Ordered ALLOW-then-BLOCK inputs exhausted budgets before BLOCK, reproducing the selection effect in fixtures rather than users. Across all 308 opportunities, including suppressions, synchronous report-call p50/p95/max was 56.157/143.356/189.324 ms. Reporting occurs before response return but after construction of the runtime elapsed field, so that field excludes reporting cost. Separate operator Health probes detected outage and recovery without demonstrating autonomous monitoring.

Answer to RQ3. Installed measurements quantify resident V3 detector activation alongside historical infrastructure costs, while leaving total and cold-start overhead unresolved. Controlled telemetry tests verify delivery and accounting, but finite privacy excess, estimation error and admission selection persist. The channel lacks performance fields, and useful deployment monitoring remains unmeasured; its content guarantee also excludes participation and network metadata.

VIII. Discussion and Limitations

A. What the System Evidence Supports

Detector placement is part of the policy decision, not simply an implementation choice. An early rule-based BLOCK prevents semantic contribution, so a better downstream model cannot repair it. Removing rules corrected 139 unnecessary restrictions but lost eighteen previously correct actions. The appropriate response is therefore to examine both restriction and protection costs, rather than infer that higher aggregate agreement justifies removal. This implication remains conditional on the provisional policy and tested fixture. A policy-correct WARN presents a related distinction: because it releases the original content, classification correctness, appropriate intervention and prevented transmission each require their own evidence.

Comparison with local sanitization clarifies who controls continuation. Casper's acknowledged warnings leave a decision to the user, while HaS and ProSan transform content to support continued use [8], [10], [11]. PriVoke instead proceeds after WARN without acknowledgement and leaves task completion unresolved after BLOCK; it neither masks prompts nor restores responses. Receiver compliance verifies that contract, but not its desirability. Establishing the latter would require evidence about authorized task completion, warning comprehension and the effort needed to reformulate a prompt.

B. Ground Truth and Generalization Remain Open

Contextual integrity concerns appropriate information flows [6], for which entity recognition supplies only part of the judgment. Independent contextual ground truth remains the largest scientific gap because PriVoke's authored targets are assistant-provisional. Freezing the separate annotation-presence endpoint before opening limits post-selection, but neither its different target nor its sole-semantic execution validates contextual authorization. Improved authored-target agreement and poor final specificity consequently answer different questions, neither establishing human agreement, appropriate intervention or task utility.

Group and text exclusions reduce declared overlap, while paired cluster analyses show sensitivity to known dependence. Neither establishes template independence, paraphrase exclusion or absence of pretraining exposure. The final endpoint comprises 308 Nemotron, 89 AI4privacy, eighty Gretel, twenty Privy and one MAPA row. These English synthetic or machine-generated cases do not represent other languages or organizational policies, and neither unadjusted intervals nor three seeds provide a population guarantee. Confirmation after opening this endpoint requires a new protected successor.

C. Retention, Availability and Practical Utility

Matching the base and exposure reduces RQ2 configuration confounding without retrospectively isolating the RQ1 package effect. Consistent with negative-flip research [7], all matched full-model candidates improved aggregate assessment yet failed the guard. Under the configured rule, higher scores cannot override casewise rejection, and no candidate was published. The weak zero-head base and small guard also constrain the significance of a pass, motivating independently reviewed retention cases and explicit criteria for harm.

Blocking when analysis is unavailable can prevent disclosure while interrupting an authorized task; WARN avoids that interruption but proceeds without informed acknowledgement. Heuristic confidence cannot resolve the choice because it is not calibrated joint correctness. Installed Linux evidence now covers historical V1 enforcement and standard V3 loading and cost, yet the quality-tier benign-prompt BLOCKs demonstrate why installation alone is not accuracy validation. Authenticated providers, Windows, workers, attachments and other transports remain unverified. Without evidence on authorized task completion, unnecessary intervention, warning comprehension and reformulation effort, practical utility and human benefit remain open.

D. Measurement Boundaries and Next Steps

Matched resident arms quantify semantic and full-detector activation, but retain instrumentation and omit extension-absent and cold-load comparisons needed to establish total overhead. For reporting, RAPPOR motivates repeated-report accounting [13]; PriVoke's ledger depends on retained state and a trusted clock, and its finite bound assumes independent uniform OS bits. Accounting leaves estimation error and admission-order effects unresolved, so privacy, estimation and representative admission require separate justification. The current deployment has too few concurrent users for a live-usage usefulness study, and more users alone would not repair the missing fields. Lifetime categorical marginals and an operator Health probe cannot replace latency/error and usage-denominator instrumentation, nor has a deployment-usefulness

threshold been established. Participation, delivery selection and metadata remain outside the bound, and aggregates supply no corrected training labels.

Across the lifecycle, early restrictions can survive later model improvements, improved candidates can remain ineligible, and admitted reports can remain unrepresentative. These findings motivate separate acceptance evidence rather than validate a remedy. Contextual adjudication, broader receiver and retention checks, total-cost measurements, and user/task evidence are needed to support stronger claims.

IX. Conclusion

PriVoke integrates local prompt inspection, explicit request enforcement, version-bound guarded revision and independently randomized reporting. Its evaluation demonstrates why each mechanism needs distinct acceptance evidence. Agreement with authored contextual targets improved, while action errors and poor specificity on a separate annotation-presence task persisted. Installed probes verified supported request handling, including unchanged-content forwarding under WARN. All matched full-model candidates improved aggregate agreement but failed retention, making average gain insufficient to justify publication. Installed V3 tests measured resident activation cost while revealing an adverse quality-tier decision; controlled telemetry verified delivery and accounting without establishing live monitoring usefulness. The contribution is an inspectable lifecycle and an empirical account of its limits. Independent contextual accuracy, useful adaptation without lost protection and human benefit remain to be established.

AI Assistance

Artificial intelligence (AI) assistance supported provisional contextual labelling and manuscript drafting. GPT-6 Sol assisted implementation and controlled testing, GPT-6 Luna assisted review, and GPT-6 Astra assisted manuscript review and rewriting. These provisional labels have not been independently adjudicated as ground truth. The authors remain responsible for the manuscript and its claims.

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